Project Title
Dictionary Query Engine with Maximum Edit Distance (Fuzzy Search)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Searching large, multilingual dictionaries becomes unreliable when users make spelling mistakes, since traditional exact-match systems return no results for even minor errors. This problem is amplified in Indian-language scripts, where Unicode complexity increases the likelihood of input mistakes, creating a barrier to effective information retrieval. This project addresses the challenge by designing a Fuzzy Search System capable of retrieving relevant words despite misspelled queries.

The system is built around the Levenshtein Edit Distance algorithm, a dynamic-programming technique that measures similarity between the input query and dictionary entries, enabling accurate ranking of suggestions by closeness of match. Efficient string-processing methods are used to index and search large dictionaries quickly, while similarity-based ranking ensures the most relevant corrections appear first. The system is further optimized for scalability, allowing it to handle extensive multilingual corpora without compromising speed, ultimately delivering fast, accurate, and Unicode-aware search results even in the presence of typing errors.

Signature of the Guide